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Teorema de diferenciación de Lebesgue para funciones continuas

Teorema 1 (de diferenciación de Lebesgue para continuas). *Sea* $f \in \mathcal{C}(\mathbb{R})$

$$\implies \forall x \in \mathbb{R} : \lim_{h \to 0} \frac{1}{2h} \int_{x-h}^{x+h} f(t)\,\mathrm{d}t = f(x).$$

***Demostración*:** Sea $x \in \mathbb{R}$. Entonces, dado $\varepsilon > 0$, $\exists \delta > 0$ tal que si $|x - y| < \delta$ se tiene que $|f(x) - f(y)| < \varepsilon$. Por tanto, si $0 < h < \delta$,

$$\left| f(x) - \frac{1}{2h} \int_{x-h}^{x+h} f(t)\,\mathrm{d}t \right| \leqslant \frac{1}{2h} \int_{x-h}^{x+h} |f(x) - f(t)|\,\mathrm{d}t < \frac{1}{2h} \int_{x-h}^{x+h} \varepsilon\,\mathrm{d}t = \varepsilon.$$

Entonces, $\forall x \in \mathbb{R} : \lim_{h \to 0} \frac{1}{2h} \int_{x-h}^{x+h} f(t)\,\mathrm{d}t = f(x)$. ■

Referenciado en

- `lem-diferenciacion-lebesgue-l1-imp-casi-toda-parte`

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